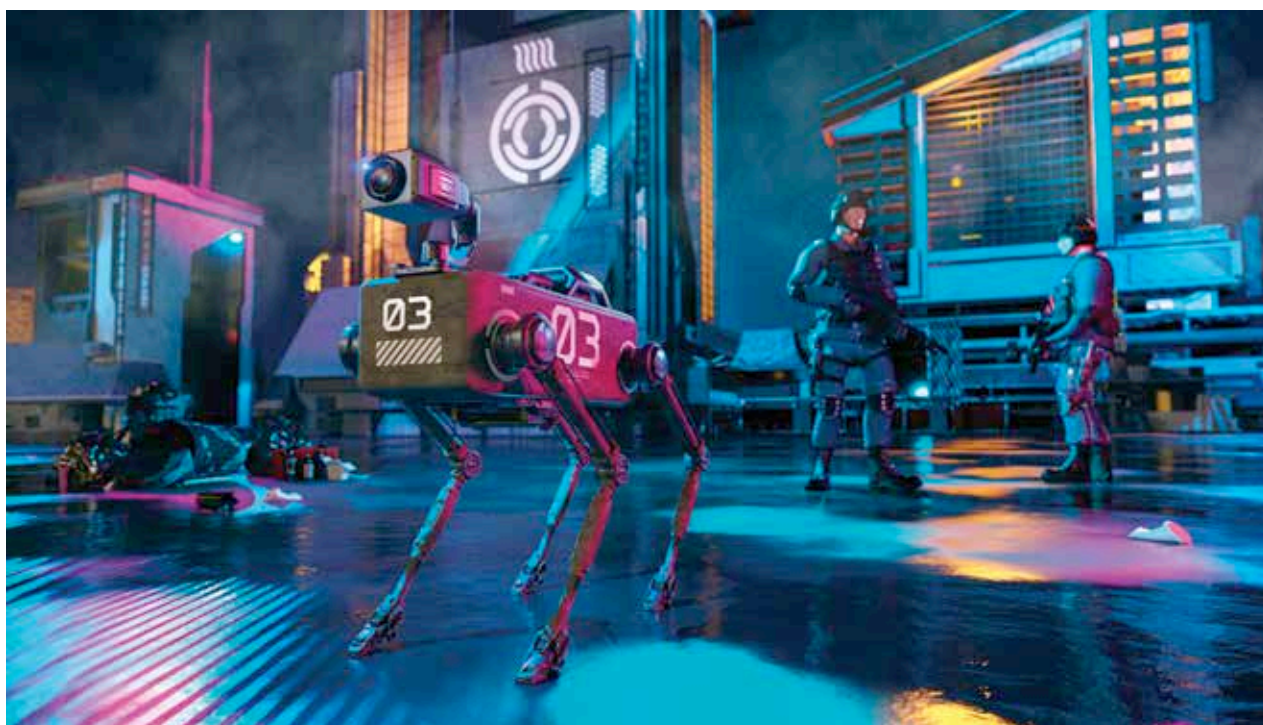




This imaginary robotic K9 unit will not only chase down criminals, it will record every moment of the chase for the judge to giggle at

tion service, with the number of cameras, days of cloud storage, and video quality scaling with the tiers. The top tier service gives you 20 wireless home security cameras (if you need all of them) with 60 days of video stored in the cloud at 4K resolution. The Wyze cameras are competitively priced and offer 14 days of free storage, but cripplingly do not allow users to upgrade to a paid tier. The D-Link security cameras offer encrypted storage in the cloud. Why cloud storage integration is not widespread among other camera categories is a mystery, and we anticipate that the capability will be essential for drones, action cameras and 360 degree cameras going

forward, just for how practical it is. Right now, the imagery captured by 360 degree cameras have to be processed by a computer and then uploaded to YouTube, which is kind of painstaking.

EVERYTHING IN SIGHT

They say the GoPro is dying, but the form factor and the practical applications means that the action camera is here to stay, at least as the most accessible security camera for bikes. The natural evolution for this category is to capture both the action and the person, with a lens on the front and the back, which results in 360 degree videos. The appropriately named GoPro Max does exactly this, but so do a whole bunch of others including the Nikon KeyMission 360, the Kodak Pixpro Orbit360 4K, the Garmin Virb 360 and the Elephone Elecam 360 demonstrating that this is a form factor that multiple manufacturers are gravitating towards.

For those looking to produce high quality 3D 360 degree videos that can be viewed in XR gear, there are offerings that keep pushing the limits in terms of resolution, and it looks like that trend will continue beyond 8K. If the videos are stitched by the

camera itself, like the Insta360 Pro does, then that is a big bonus. The Insta360 Titan is leading in the space providing 3D 11K footage along with impressive low light performance. The Suometry OmniPolar 360 platform uses four cameras to capture 3D VR video, along with a depth map of the surroundings, with the differentiator being the ability to fine tune the stereo spacing. The platform too provides a prestitched video ready for editing. These pioneering models serve as an indication of things to come.

These action cameras and even the drones will require dedicated chips for computer vision, and right now the leading SoC manufacturer is Ambarella, which has made chips for GoPro, DJI, Xiaomi, and competition is scarce at the moment but the camera industry is highly vertically integrated. The latest CV5 chip announced in January this year can support one 8K stream or four simultaneous 4K streams, with an AI engine geared towards automotive and robotic applications.

THE RETRO COMEBACK

One of the biggest draws of photography is the nostalgia factor, and there is no better way to capture



The Canon Powershot Zoom